Lecture 4: Shell Scripting and Package Managers

Objective:
Gain proficiency in shell scripting and package management for High-Performance Computing (HPC) system setup.

Duration: 1 Hour

🎯 Learning Outcomes

By the end of this lecture, students will be able to:

- Write basic shell scripts using Bash
- Automate common tasks using loops and conditionals
- Use Linux package managers (apt, yum, dnf) to install and manage software
- Understand the relevance of these tools in HPC environments

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🧩 1. Introduction to Shell Scripting

Shell scripting is a method to automate tasks in Linux environments using a sequence of shell commands written in a file.

Common Shells:

- **Bash** (Bourne Again SHell) – most widely used
- Others: zsh, sh, csh

🔤 2. Bash Basics

Creating a shell script:

```
#!/bin/bash
echo "Hello, HPC World!"
```

- Save as `script.sh`

- Make executable: `chmod +x script.sh`
- Run: `./script.sh`

🔧 3. Variables in Bash

```
name="Ram"
echo "Welcome, $name"
```

🔁 4. Loops and Conditionals

For Loop Example:

```
for i in {1..5}
do
   echo "Iteration $i"
done
```

If Condition Example:

```
if [ $age -gt 18 ]
then
  echo "Adult"
else
  echo "Minor"
fi
```

⚙️ 5. Automating HPC Setup with Scripts

Example: Script to install essential packages

```
#!/bin/bash
sudo apt update
sudo apt install -y build-essential htop git
```

Use cases in HPC:

- Automating node setup
- Job scheduling script
- Monitoring logs

📦 6. Package Managers Overview

Package managers are tools that automate the process of installing, upgrading, configuring, and removing software packages.

🔹 APT (Debian/Ubuntu)

- `sudo apt update` – Updates package index
- `sudo apt install gcc` – Installs GCC compiler
- `sudo apt remove nano` – Removes nano editor

🔹 YUM/DNF (RHEL/CentOS/Fedora)

- `sudo yum install wget` – (YUM command)
- `sudo dnf install wget` – (DNF, modern YUM)
- `sudo yum remove wget`

Use in HPC:

- Installing compilers, MPI libraries, job schedulers like SLURM

💡 Real-World HPC Examples

- Automating script to set up MPI across cluster nodes
- Shell script to submit and manage SLURM jobs
- Package manager used to install TensorFlow in a compute node

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🏫 Classroom Activities

Activity 1: Write Your First Shell Script

- Task: Create a script that prints your name, current date, and list of files in home directory
- Time: 10 minutes

Activity 2: Simulate Software Installation

- Instructor demonstrates `apt install`, `yum remove`
- Students simulate with dry-run or local VM

Activity 3: Group Discussion

- Topic: Why scripting is critical in HPC environments?
- Outcome: Identify 3 tasks that can be automated

📚 Summary

- Shell scripting is vital for automating repetitive admin and research tasks in HPC.
- Bash is the most commonly used shell.
- Package managers like **apt**, **yum**, and **dnf** streamline software installation and updates.
- These skills are foundational for maintaining and scaling HPC systems.

📝 Assignment

- Write a shell script that:
    1. Updates the system
    2. Installs Git and Python
    3. Creates a new directory named `hpc_tools`
    4. Redirects output of `top` command to a file
- Submit via LMS before next lecture.